

STANES SCHOOL

1044, Avinashi Road, Coimbatore – 18.

UNIT TEST – 1 2026-2027

Class : XI
Subject : Computer Science

Duration : 2 Hours
Max. Marks : 50

SECTION A (17 × 1 = 17 Marks)

- State whether the following statement is True or False:
Python is an interpreted, high-level programming language created by Guido van Rossum and first released in 1991.
- Which of the following is a **mutable** data type in Python?
a) tuple b) str c) int d) list
- What will be the output of the following code?
print(type(True))
a) <class 'int'> b) <class 'bool'> c) <class 'str'> d) Error
- Which of the following is **not** a valid Python keyword?
a) elif b) func c) None d) pass
- Which of the following is a **valid identifier** in Python?
a) 2ndYear b) my-var c) _total_marks d) class
- Write the output of the following Python code:
x = 17
y = 5
print(x // y, x % y, x ** 2)
- What will be the output of the following Python statement?
print(4 + 3 ** 2 - 12 // 4)
- Which of the following operators checks if two variables refer to the same object in memory?
a) == b) != c) is d) in
- Identify the correct statement:
a) Python is case-insensitive b) Python is dynamically typed
c) Python requires variable type declaration d) Python does not support Unicode
- State True or False:
input() always returns data as a string.
- What will be the output of the following Python code?
a = b = c = 50
print(a, b, c)
a) 50 50 50 b) 50 c) Error d) 50 50
- What will be the output of the following Python code?
fruits = ['apple', 'banana', 'mango']
print('grape' not in fruits)
a) False b) True c) Error d) None
- What will be the output of the following Python code?
x = 15
print(x / 4)
print(x // 4)
a) 3.75 and 3 b) 3 and 3.75 c) 3.75 and 4 d) Error
- Assertion (A):** In Python, a list can be used as a key in a dictionary.
Reason (R): Only immutable data types (int, float, str, tuple, bool) are allowed as dictionary keys.
Choose the correct option:
a) Both A and R are True and R is the correct explanation for A
b) Both A and R are True but R is not the correct explanation for A
c) A is True but R is False
d) A is False but R is True
- Which of the following is used to take input from the user in Python?
a) cin() b) scan() c) input() d) read()
- State whether the following statement is True or False:
After using the **del** keyword to delete a variable, only the value is deleted and the variable can still be accessed in the program.
- What will be the output of the following Python code?
print(not False and True or False)
a) True b) False c) Error d) None

SECTION B (5 × 2 = 10 Marks)

- Explain the difference between **implicit type conversion** and **explicit type conversion** in Python with a suitable example of each.

19. The code below is intended to accept a student's name and marks from the user and display them. Rewrite it after correcting **all errors**. Underline the corrections made.

```
n = Input('Enter name: ')
m = int(input('Enter marks: '))
Print('Name:', n)
Print('Marks:' marks)
```

20. (Using Python built-in functions only):

- I. Write a statement to convert the integer 255 to a string and store it in variable s.
- II. Write a statement to convert the string 'True' to a boolean value.

21. Write the output of the following codes:

```
a. f=2
   g=f+3
   f+=g
   print(f,g)
b. w=14
   t=4
   print(w%3**2*t)
```

22. What is the output of the following code:

```
p = False
q = True
r = False
print (p or (q and r))
print ((p or q) and (not r))
```

SECTION C (2 × 3 = 6 Marks)

23. Evaluate the following Python expression step-by-step and write the final output. Show each step clearly.

```
result = 30 - 2 ** 3 ** 2 // 4 * 3 + 5
print(result)
```

24. Write a Python program to calculate the area of a rectangle. The program should:

1. Ask the user to input the length and breadth of the rectangle.
2. Calculate the area (area = length * breadth).
3. Display the area with an appropriate message.

SECTION D (3 × 4 = 12 Marks)

25. Write a Python program to convert temperature from Celsius to Fahrenheit.

The formula for conversion is $F = (C * 9/5) + 32$. The program should:

1. Prompt the user to enter the temperature in Celsius.
2. Calculate the temperature in Fahrenheit.
3. Display the result.

26. Write codes to perform the following:

1. Create a dictionary with the names of 5 students as keys and their marks as values.
2. Code that takes a string from the user and prints the string and its length.
3. Create a tuple that stores the days of a week.
4. Create a list of your favourite cars.

27. Write a code to calculate BMI (Body Mass Index) of a person. Body Mass Index is a simple calculation using a person's height and weight. The formula is **BMI=kg/m²**, where kg is a person's weight in kilograms and m² is their height in metres squared.

SECTION E (1 × 5 = 5 Marks)

28. List and explain the five categories of tokens in Python. Also, categorize the following into appropriate tokens:

1. False 2. % 3. continue 4. name 5. (6. > 7. : 8. value